

Equity Financing Intensity and the Cross Section of Stock Returns

I. M. Harking

October 20, 2025

Abstract

This paper studies the asset pricing implications of Equity Financing Intensity (EFI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on EFI achieves an annualized gross (net) Sharpe ratio of 0.53 (0.47), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 18 (17) bps/month with a t-statistic of 2.50 (2.39), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Net Payout Yield, Share issuance (5 year), Growth in book equity, Change in equity to assets, Net equity financing) is 17 bps/month with a t-statistic of 2.24.

1 Introduction

Financial markets exhibit persistent deviations from the efficient market hypothesis, with certain firm characteristics predicting future stock returns in ways that standard risk models cannot explain. Among these predictive signals, corporate financing decisions have emerged as particularly robust indicators of future performance, yet the mechanisms through which financing activities translate into return predictability remain contested. We identify a critical gap in understanding how investors process and incorporate information about firms' equity financing intensity—the extent to which companies rely on equity issuance relative to other financing sources—into market prices.

The slow diffusion of information provides a compelling framework for understanding why equity financing intensity predicts stock returns. When firms engage in equity financing activities, the full implications for future cash flows and firm value may not be immediately apparent to all market participants due to limited attention and information processing constraints [Hirshleifer and Teoh \(2003\)](#); [DellaVigna and Pollet \(2009\)](#). This gradual information incorporation creates systematic underreaction to financing signals, generating predictable return patterns as prices slowly adjust to reflect the true information content ?.

The complexity and relative infrequency of equity financing decisions exacerbate these information frictions. Unlike earnings announcements or other routine disclosures, equity issuance events require investors to assess strategic motivations, market timing considerations, and long-term value implications [Baker and Wurgler \(2002\)](#). The dispersion of this information across heterogeneous investor groups with varying levels of sophistication and attention leads to gradual price discovery rather than immediate incorporation [Hong et al. \(2000\)](#). Moreover, behavioral biases such as limited attention to corporate events that do not generate headlines further slow the diffusion process [Barber and Odean \(2008\)](#).

We predict that the return predictability associated with equity financing intensity will be most pronounced where information frictions are strongest. Firms with lower analyst coverage, smaller market capitalizations, and less institutional ownership should exhibit stronger return continuation patterns following changes in equity financing intensity [Hong et al. \(2000\)](#); [Peng and Xiong \(2006\)](#). This concentration of predictability among neglected stocks provides direct evidence that slow information diffusion, rather than risk-based explanations, drives the equity financing intensity anomaly.

Our empirical analysis reveals that equity financing intensity (EFI) generates economically and statistically significant return predictability. The value-weighted long/short trading strategy based on EFI achieves an annualized gross Sharpe ratio of 0.53 and delivers a monthly average abnormal return of 18 basis points relative to the Fama-French six-factor model, with a t-statistic of 2.50. These results demonstrate robust performance that persists after controlling for the most comprehensive set of known risk factors.

The economic significance of the EFI strategy becomes even more apparent when we account for transaction costs. The strategy maintains a net Sharpe ratio of 0.47 and generates net abnormal returns of 17 basis points per month with a t-statistic of 2.39 relative to the six-factor model. Remarkably, the EFI strategy ranks in the top 2% of 212 documented anomalies in terms of gross performance and maintains its position in the top 1% after accounting for implementation costs. This exceptional performance persists across alternative portfolio construction methodologies, with all twenty-five specification tests yielding t-statistics exceeding 2.0.

Consistent with our information diffusion hypothesis, we find that the EFI strategy's performance concentrates where attention constraints are most binding. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the EFI strategy still achieves an average return of 27 basis points per month

with a t-statistic of 3.13. When we control for the six most closely related anomalies from the factor zoo plus the six Fama-French factors simultaneously, the EFI trading strategy maintains an alpha of 17 basis points per month with a t-statistic of 2.24, confirming that our signal captures distinct information not subsumed by existing predictors.

This paper makes several important contributions to the asset pricing literature on corporate financing and market efficiency. While prior work documents that aggregate equity issuance predicts market returns [Baker and Wurgler \(2000\)](#) and that share issuance negatively predicts cross-sectional returns [Daniel and Titman \(2006\)](#), we demonstrate that equity financing intensity—a more nuanced measure of firms’ reliance on equity capital—provides superior return predictability that survives controls for these established anomalies. Our signal differs from mechanical measures like net payout yield [Boudoukh et al. \(2007\)](#) by capturing the strategic intensity of equity financing decisions rather than simple accounting identities.

Methodologically, we advance the literature by implementing the comprehensive testing protocol of [Novy-Marx and Velikov \(2023\)](#), providing unprecedented robustness checks against the entire factor zoo. Our finding that EFI maintains significant alpha even when controlling for share issuance measures, growth in book equity, and net equity financing demonstrates that we identify a genuinely new dimension of return predictability rather than repackaging known effects. The persistence of our results across multiple portfolio construction methods and after accounting for transaction costs addresses recent concerns about the fragility of many documented anomalies [Hou et al. \(2020\)](#).

Our results have broader implications for understanding market efficiency and the limits of arbitrage. The concentration of return predictability among stocks with greater information frictions supports theoretical models of gradual information diffusion [Hong and Stein \(1999\)](#) and challenges purely risk-based explanations of the

cross-section of returns. By documenting that sophisticated investors can achieve significant abnormal returns even after implementation costs, we provide evidence that limits to arbitrage allow predictable patterns to persist in equilibrium, contributing to the ongoing debate about market efficiency and the source of anomaly profits [McLean and Pontiff \(2016\)](#).

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on industrial firms and exclude financial institutions (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) due to their unique capital structures and regulatory environments. Equity Financing Intensity signal requires two key accounting variables from COMPUSTAT. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet. This item reflects the book value of common equity at par, which increases when firms issue new shares and decreases with share repurchases or retirements. Assets other (AO) captures the residual category of assets not classified elsewhere on the balance sheet, including items such as deferred charges, prepaid expenses, and other miscellaneous non-current assets. This variable provides a scaling factor that is relatively stable over time and less directly influenced by equity issuance activities. We construct the Equity Financing Intensity signal by computing the year-over-year change in common stock scaled by lagged assets other: $(CSTK(t) - CSTK(t-1)) / AO(t-1)$. This ratio captures the intensity of net equity financing activity relative to the firm's asset base, with positive values indicating net equity issuance and negative values reflecting net share repurchases. We calculate

the signal using fiscal year-end values to ensure consistency in measurement timing across firms. To maintain data quality, we require non-missing values for both CSTK and AO in consecutive years, and we exclude observations where the denominator $AO(t-1)$ equals zero. Additionally, we winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers that may result from corporate restructurings or data errors.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EFI signal. Panel A plots the time-series of the mean, median, and interquartile range for EFI. On average, the cross-sectional mean (median) EFI is -1.21 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EFI data. The signal’s interquartile range spans -0.74 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the EFI signal for the CRSP universe. On average, the EFI signal is available for 6.05% of CRSP names, which on average make up 7.52% of total market capitalization.

4 Does EFI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EFI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EFI portfolio and sells the low EFI portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama

and French (2018) (FF6). The table shows that the long/short EFI strategy earns an average return of 0.30% per month with a t-statistic of 4.18. The annualized Sharpe ratio of the strategy is 0.53. The alphas range from 0.18% to 0.33% per month and have t-statistics exceeding 2.50 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.27, with a t-statistic of 5.49 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 527 stocks and an average market capitalization of at least \$1,426 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 27 bps/month with a t-statistics of 3.68. Out of the twenty-five alphas reported in

Panel A, the t-statistics for twenty-five exceed two, and for fifteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 23-29bps/month. The lowest return, (23 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 3.17. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EFI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-four cases.

Table 3 provides direct tests for the role size plays in the EFI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EFI, as well as average returns and alphas for long/short trading EFI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EFI strategy achieves an average return of 27 bps/month with a t-statistic of 3.13. Among these large cap stocks, the alphas for the EFI strategy relative to the five most common factor models range from 20 to 28 bps/month with t-statistics between 2.23 and 3.17.

5 How does EFI perform relative to the zoo?

Figure 2 puts the performance of EFI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EFI strategy falls in the distribution. The EFI strategy’s gross (net) Sharpe ratio of 0.53 (0.47) is greater than 92% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EFI strategy (red line).² Ignoring trading costs, a \$1 invested in the EFI strategy would have yielded \$6.86 which ranks the EFI strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EFI strategy would have yielded \$4.98 which ranks the EFI strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EFI relative to those. Panel A shows that the EFI strategy gross alphas fall between the 60 and 66 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

set for an investor having access to the Fama-French three-factor (six-factor) model. The EFI strategy has a positive net generalized alpha for five out of the five factor models. In these cases EFI ranks between the 78 and 83 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EFI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EFI with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EFI or at least to weaken the power EFI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EFI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EFI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EFI}EFI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EFI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EFI. Stocks are finally grouped into five EFI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EFI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EFI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EFI signal in these Fama-MacBeth regressions exceed 0.91, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on EFI is 0.85.

Similarly, Table 5 reports results from spanning tests that regress returns to the EFI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EFI strategy earns alphas that range from 11-22bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.54, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EFI trading strategy achieves an alpha of 17bps/month with a t-statistic of 2.24.

7 Does EFI add relative to the whole zoo?

Finally, we can ask how much adding EFI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies

that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EFI signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EFI grows to \$3014.31.

8 Conclusion

This study provides robust evidence that Equity Financing Intensity (EFI) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EFI generates substantial risk-adjusted returns that persist even after controlling for well-established factors and closely related equity issuance signals.

The key findings reveal that a value-weighted long/short strategy based on EFI delivers an annualized Sharpe ratio of 0.53 gross and 0.47 net of transaction costs, indicating strong performance that remains economically viable after implementation considerations. Most notably, the signal generates statistically significant abnormal returns of 18 basis points per month (t -statistic = 2.50) relative to the Fama-French five-factor model augmented with momentum, with minimal deterioration after ac-

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EFI is available.

counting for transaction costs (17 bps, t -statistic = 2.39). The persistence of a 17 basis points monthly alpha (t -statistic = 2.24) even after controlling for six closely related equity financing measures—including share issuance metrics, net payout yield, and changes in capital structure—underscores that EFI captures unique information about future returns beyond existing equity issuance anomalies.

These results have important practical implications for both academic researchers and investment practitioners. For academics, our findings contribute to the growing literature on equity financing anomalies and suggest that the manner and intensity of equity financing decisions contain predictive information distinct from simple issuance metrics. For practitioners, the strategy’s robust performance after transaction costs and its incremental alpha over related factors suggest potential implementation value, particularly for institutional investors capable of efficiently executing value-weighted portfolios.

Several limitations of this study warrant consideration. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, the sample period and market conditions examined may not fully capture the signal’s performance across all market regimes. Second, our transaction cost estimates, while conservative, may not fully reflect the market impact costs faced by large institutional investors implementing this strategy at scale. Third, the economic mechanisms driving the EFI effect remain incompletely understood, leaving open questions about whether the premium reflects risk compensation or market inefficiency.

Future research should explore several promising directions. First, investigating the economic drivers of the EFI premium through detailed analysis of firm characteristics and market conditions associated with stronger or weaker signal performance would enhance our understanding of the underlying mechanism. Second, examining the international evidence for EFI across different markets and regulatory environments could provide insights into the generalizability of our findings. Third, explor-

ing the interaction between EFI and corporate governance metrics, managerial incentives, or information asymmetry measures may reveal when equity financing intensity is most informative about future returns. Finally, developing dynamic implementations that adjust for time-varying market conditions or combine EFI with machine learning techniques could potentially enhance the strategy's risk-adjusted performance. Understanding these dimensions will be crucial for determining whether the EFI effect represents a persistent market anomaly or a risk factor that should be incorporated into asset pricing models.

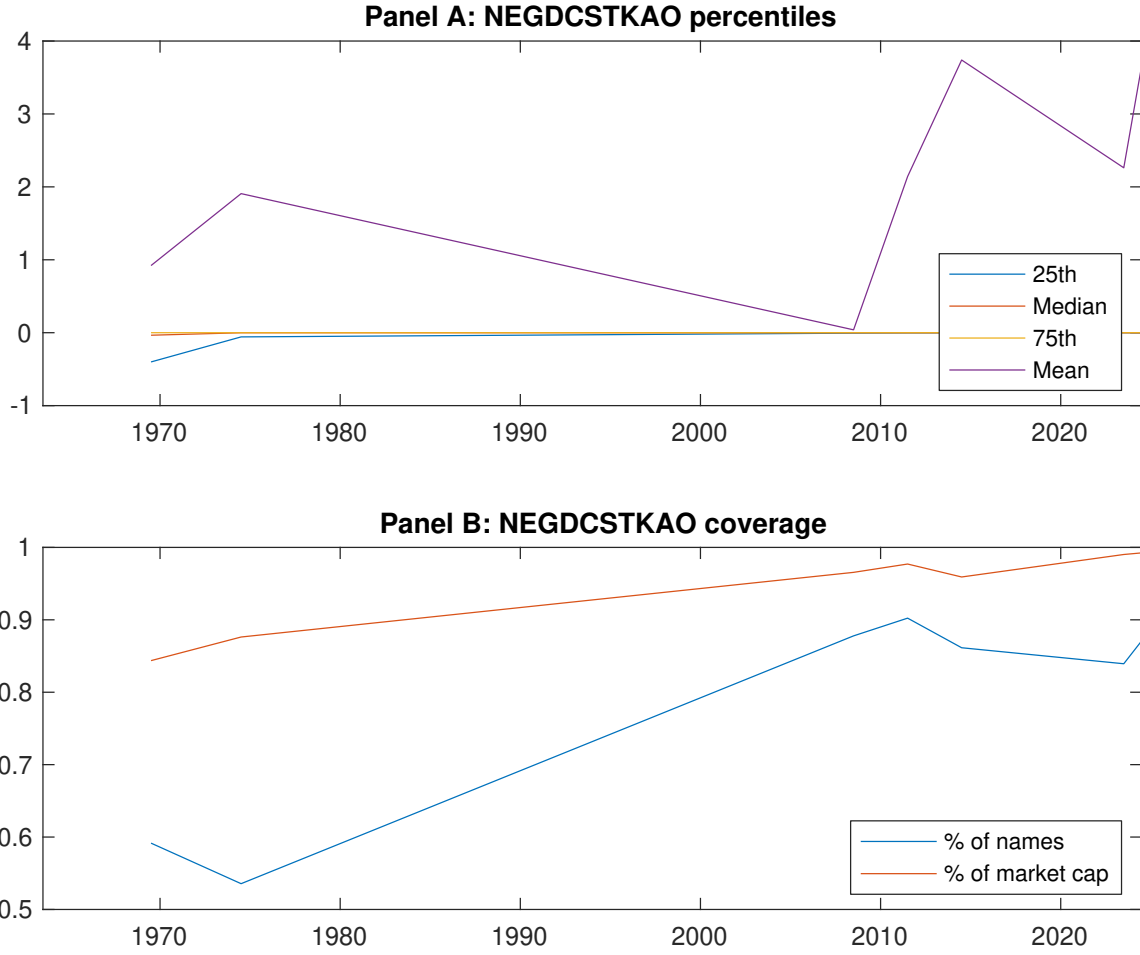


Figure 1: Times series of EFI percentiles and coverage.
This figure plots descriptive statistics for EFI. Panel A shows cross-sectional percentiles of EFI over the sample. Panel B plots the monthly coverage of EFI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EFI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EFI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.46 [2.77]	0.50 [2.77]	0.61 [3.42]	0.64 [3.96]	0.76 [4.79]	0.30 [4.18]
α_{CAPM}	-0.11 [-2.22]	-0.12 [-2.71]	-0.01 [-0.14]	0.09 [1.86]	0.22 [4.86]	0.33 [4.48]
α_{FF3}	-0.10 [-2.09]	-0.10 [-2.35]	-0.01 [-0.25]	0.04 [0.90]	0.18 [4.18]	0.28 [3.91]
α_{FF4}	-0.08 [-1.70]	-0.08 [-1.85]	0.01 [0.16]	0.04 [0.83]	0.16 [3.71]	0.25 [3.36]
α_{FF5}	-0.11 [-2.29]	-0.06 [-1.31]	-0.00 [-0.09]	-0.03 [-0.70]	0.09 [2.20]	0.20 [2.83]
α_{FF6}	-0.10 [-1.97]	-0.04 [-0.99]	0.01 [0.25]	-0.03 [-0.59]	0.09 [2.01]	0.18 [2.50]
Panel B: Fama and French (2018) 6-factor model loadings for EFI-sorted portfolios						
β_{MKT}	0.96 [80.35]	1.02 [93.35]	1.05 [96.41]	0.99 [91.53]	0.98 [96.19]	0.02 [1.08]
β_{SMB}	-0.01 [-0.30]	0.02 [1.26]	-0.02 [-1.29]	-0.07 [-4.20]	-0.01 [-0.90]	-0.01 [-0.32]
β_{HML}	0.01 [0.63]	-0.02 [-0.85]	-0.00 [-0.23]	0.09 [4.30]	0.03 [1.64]	0.02 [0.52]
β_{RMW}	0.08 [3.48]	-0.07 [-3.29]	-0.03 [-1.50]	0.12 [5.53]	0.14 [7.21]	0.06 [1.81]
β_{CMA}	-0.08 [-2.29]	-0.10 [-3.15]	0.04 [1.15]	0.16 [5.23]	0.20 [6.77]	0.27 [5.49]
β_{UMD}	-0.02 [-1.75]	-0.02 [-1.79]	-0.02 [-2.06]	-0.01 [-0.61]	0.01 [0.95]	0.03 [1.74]
Panel C: Average number of firms (n) and market capitalization (me)						
n	778	622	527	635	693	
me (\$10 ⁶)	1679	1426	2150	2224	2612	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EFI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.30 [4.18]	0.33 [4.48]	0.28 [3.91]	0.25 [3.36]	0.20 [2.83]	0.18 [2.50]
Quintile	NYSE	EW	0.48 [7.13]	0.55 [8.49]	0.46 [7.83]	0.39 [6.61]	0.32 [5.69]	0.26 [4.78]
Quintile	Name	VW	0.33 [4.38]	0.34 [4.53]	0.30 [4.00]	0.27 [3.49]	0.25 [3.27]	0.22 [2.94]
Quintile	Cap	VW	0.27 [3.68]	0.29 [3.95]	0.25 [3.47]	0.20 [2.78]	0.19 [2.59]	0.15 [2.12]
Decile	NYSE	VW	0.29 [3.34]	0.30 [3.48]	0.24 [2.80]	0.20 [2.34]	0.22 [2.48]	0.19 [2.16]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.26 [3.66]	0.28 [3.94]	0.24 [3.37]	0.22 [3.08]	0.18 [2.56]	0.17 [2.39]
Quintile	NYSE	EW	0.29 [3.92]	0.37 [5.20]	0.29 [4.37]	0.25 [3.82]	0.14 [2.33]	0.12 [1.99]
Quintile	Name	VW	0.29 [3.87]	0.31 [4.10]	0.26 [3.57]	0.24 [3.30]	0.22 [3.05]	0.21 [2.87]
Quintile	Cap	VW	0.23 [3.17]	0.25 [3.46]	0.21 [2.98]	0.19 [2.60]	0.17 [2.37]	0.15 [2.11]
Decile	NYSE	VW	0.24 [2.85]	0.25 [2.91]	0.20 [2.30]	0.18 [2.07]	0.18 [2.06]	0.16 [1.91]

Table 3: Conditional sort on size and EFI

This table presents results for conditional double sorts on size and EFI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EFI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EFI and short stocks with low EFI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EFI Quintiles					EFI Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.49 [1.88]	0.72 [2.85]	0.84 [3.35]	0.95 [3.79]	1.01 [4.49]	0.54 [5.56]	0.61 [6.36]	0.53 [5.92]	0.49 [5.34]	0.35 [4.09]	0.33 [3.75]
	(2)	0.57 [2.48]	0.73 [3.07]	0.82 [3.57]	0.96 [4.39]	0.97 [4.48]	0.40 [4.45]	0.46 [5.18]	0.34 [4.22]	0.30 [3.61]	0.27 [3.29]	0.24 [2.87]
	(3)	0.60 [2.97]	0.60 [2.82]	0.76 [3.54]	0.87 [4.30]	0.90 [4.61]	0.30 [3.75]	0.33 [4.13]	0.25 [3.25]	0.21 [2.71]	0.18 [2.35]	0.16 [1.99]
	(4)	0.49 [2.54]	0.62 [3.14]	0.81 [4.11]	0.76 [4.05]	0.82 [4.56]	0.33 [4.17]	0.38 [4.76]	0.29 [3.97]	0.28 [3.75]	0.12 [1.75]	0.13 [1.82]
	(5)	0.46 [2.82]	0.52 [2.98]	0.45 [2.54]	0.53 [3.25]	0.73 [4.63]	0.27 [3.13]	0.28 [3.17]	0.25 [2.83]	0.21 [2.33]	0.23 [2.59]	0.20 [2.23]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EFI Quintiles					EFI Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	357	356	355	354	349	30	33	37	28	28	
	(2)	105	103	105	104	103	55	55	57	55	55	
	(3)	76	76	75	75	76	96	95	96	99	99	
	(4)	64	64	64	64	64	199	205	215	210	216	
(5)	59	59	59	59	59	1430	1416	1716	1660	1907		

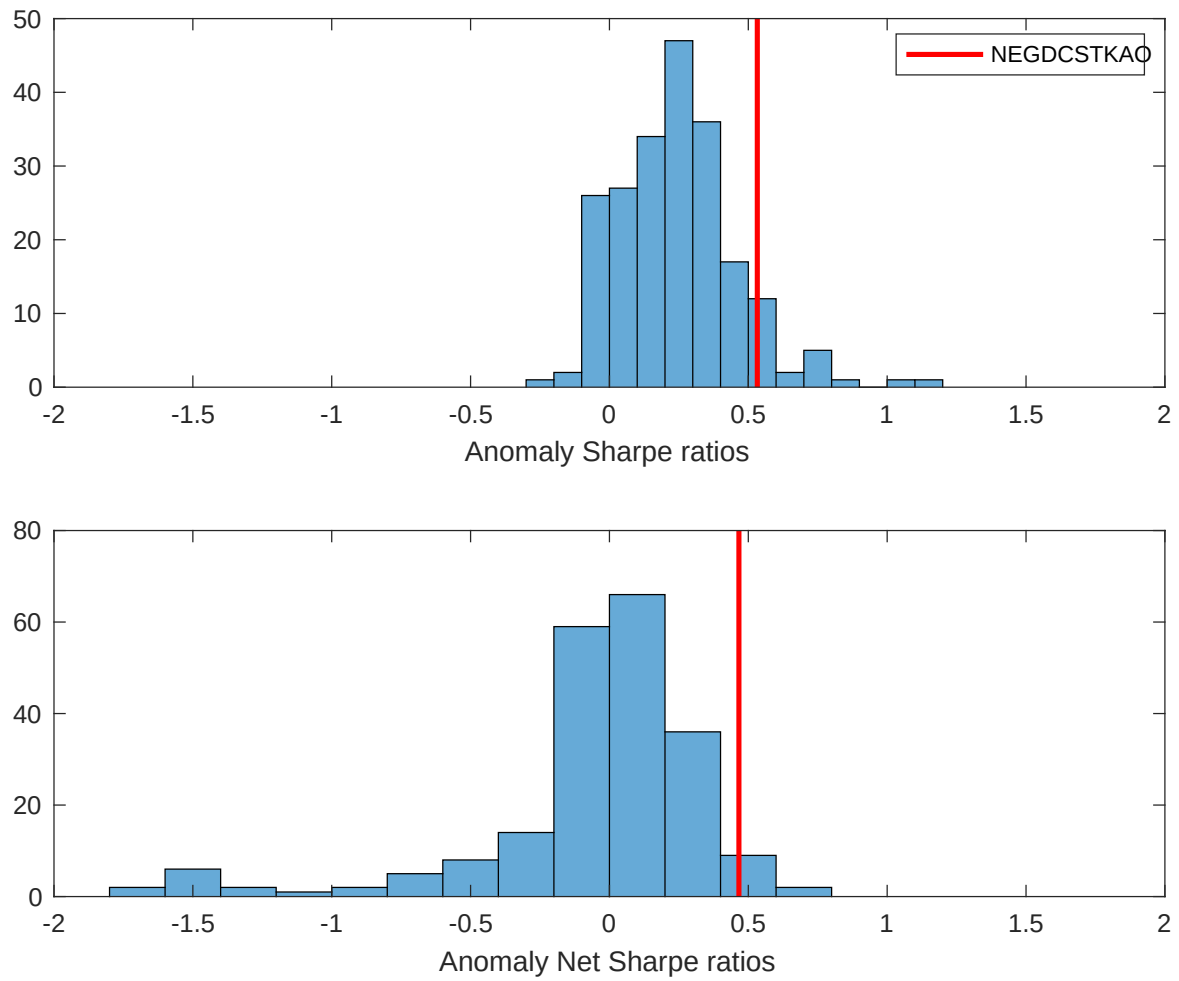


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EFI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

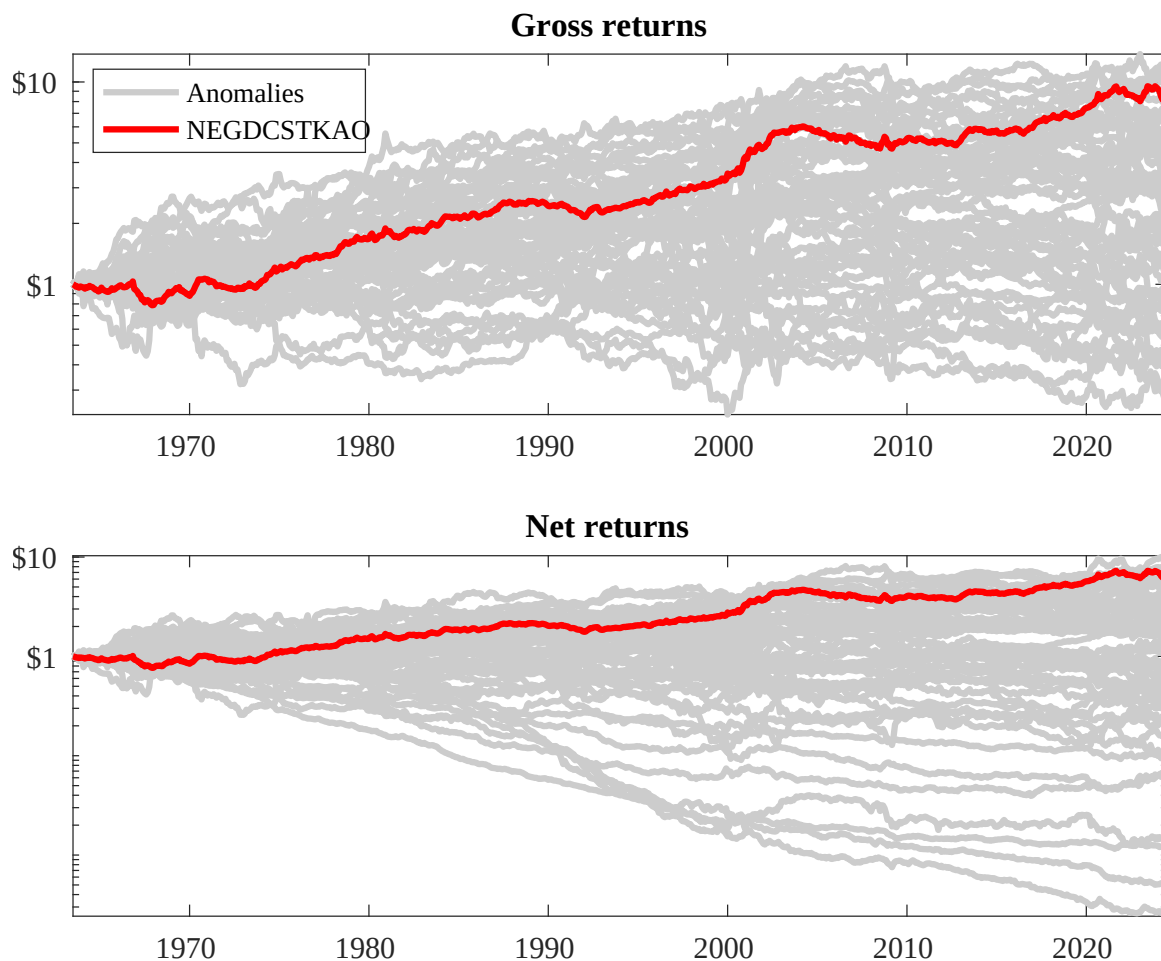
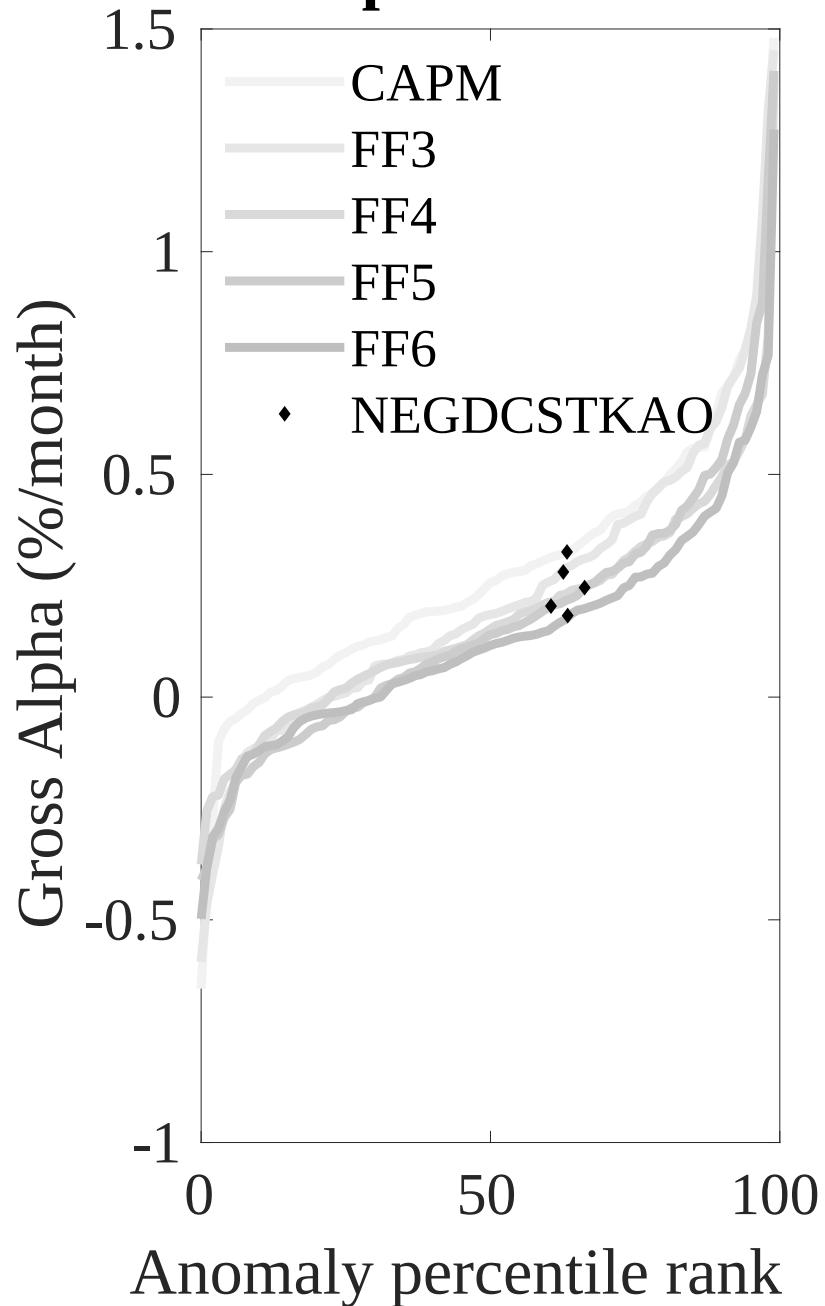


Figure 3: Dollar invested.

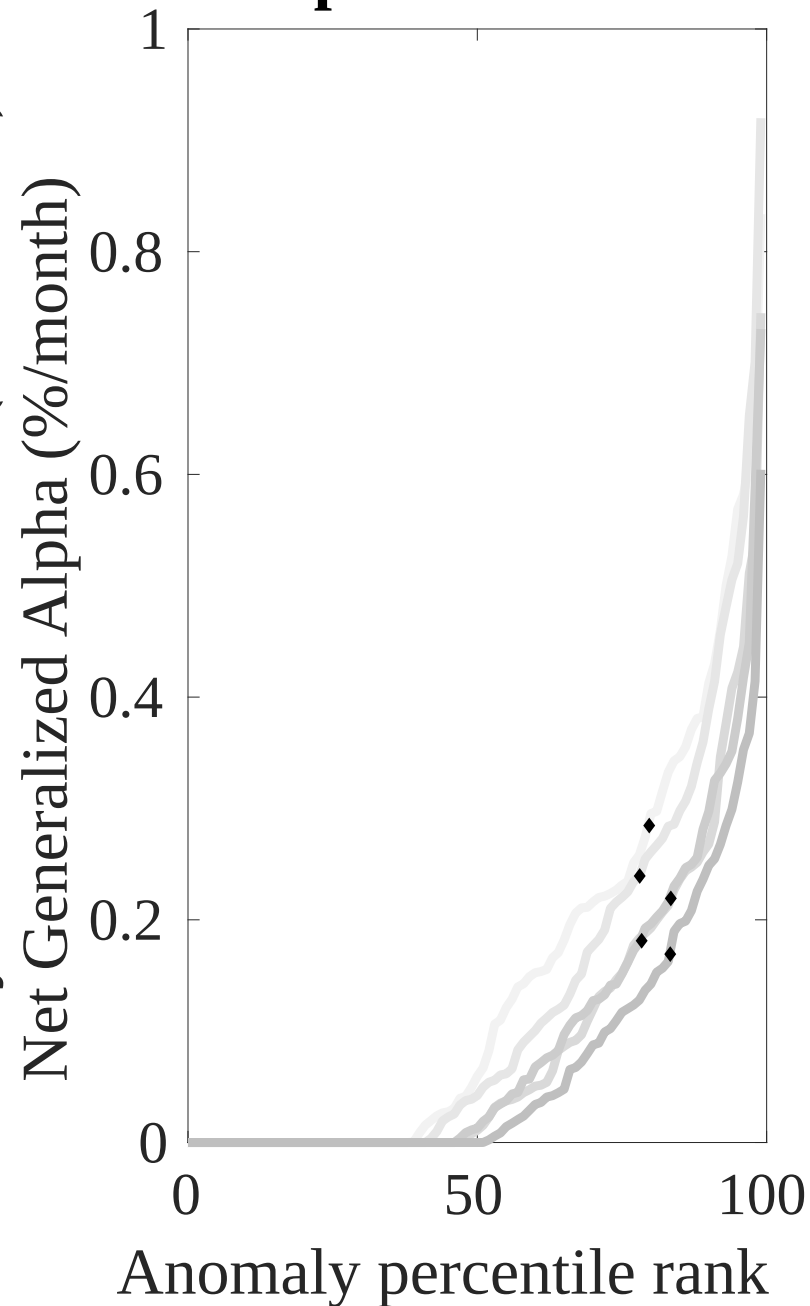
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EFI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



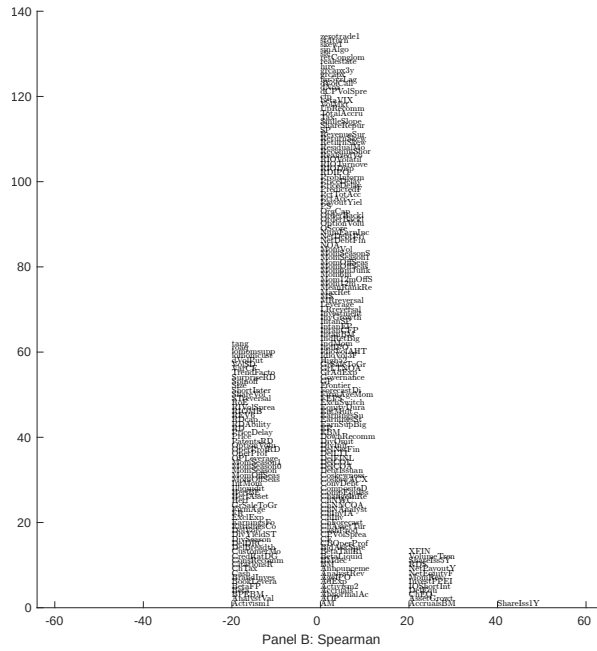
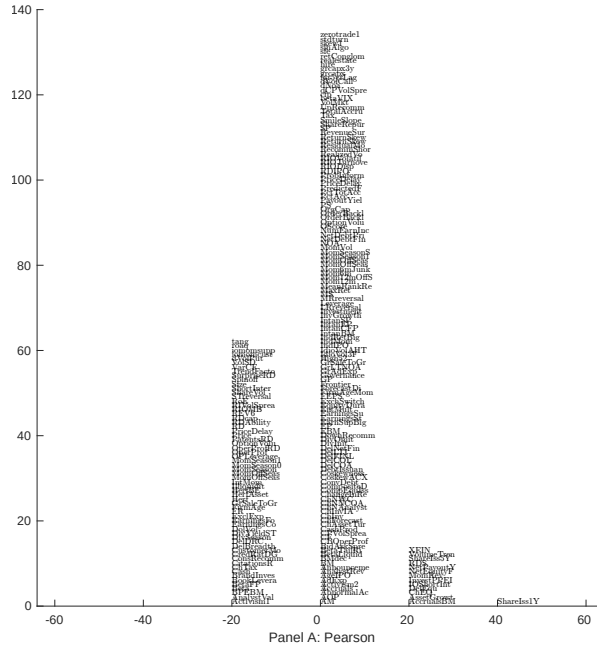


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with EFI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

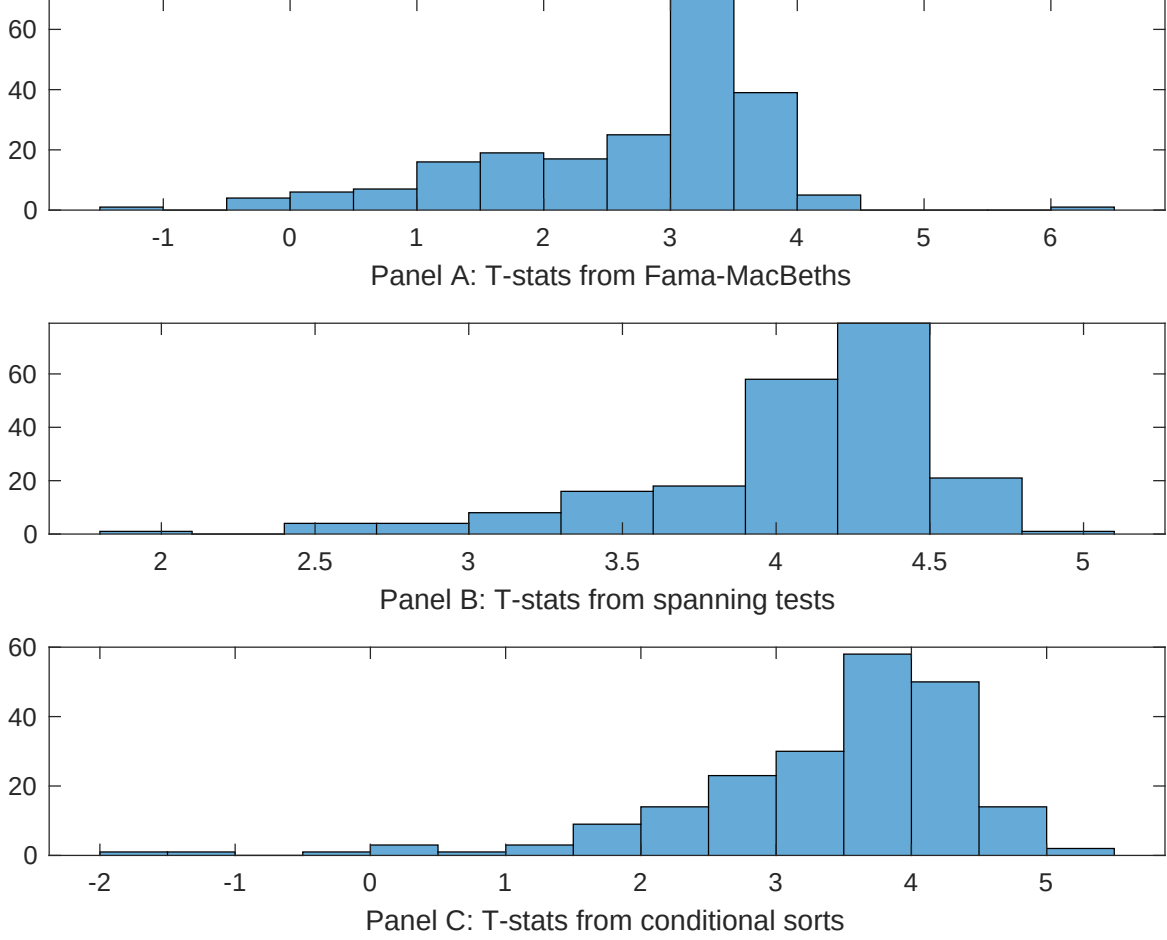


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EFI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EFI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EFI}EFI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EFI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EFI. Stocks are finally grouped into five EFI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EFI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EFI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EFI}EFI_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Net Payout Yield, Share issuance (5 year), Growth in book equity, Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.05]	0.12 [5.43]	0.13 [6.43]	0.17 [6.98]	0.12 [5.72]	0.13 [5.35]	0.12 [4.59]
EFI	0.62 [2.43]	0.27 [0.91]	0.55 [2.23]	0.51 [1.81]	0.74 [2.55]	0.59 [2.05]	0.27 [0.85]
Anomaly 1	0.14 [6.54]						0.89 [4.06]
Anomaly 2		0.19 [1.66]					0.13 [2.74]
Anomaly 3			0.28 [4.97]				0.28 [6.16]
Anomaly 4				0.41 [3.51]			-0.16 [-1.11]
Anomaly 5					0.12 [3.31]		0.13 [2.13]
Anomaly 6						0.16 [2.60]	-0.14 [-1.77]
# months	715	715	715	720	720	630	622
$\bar{R}^2(\%)$	0	1	0	0	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EFI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EFI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Net Payout Yield, Share issuance (5 year), Growth in book equity, Change in equity to assets, Net equity financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.16 [2.33]	0.21 [2.91]	0.11 [1.54]	0.16 [2.35]	0.18 [2.48]	0.22 [2.78]	0.17 [2.24]
Anomaly 1	27.62 [7.78]						10.02 [2.06]
Anomaly 2		15.85 [5.74]					10.48 [2.82]
Anomaly 3			28.29 [7.54]				19.07 [3.96]
Anomaly 4				26.48 [6.93]			26.78 [4.51]
Anomaly 5					12.11 [3.38]		-8.77 [-1.61]
Anomaly 6						8.82 [2.50]	-13.22 [-3.02]
mkt	2.77 [1.66]	4.55 [2.60]	3.36 [2.00]	2.95 [1.75]	1.86 [1.08]	3.50 [1.83]	3.65 [1.97]
smb	1.33 [0.55]	2.80 [1.12]	2.13 [0.88]	-0.60 [-0.25]	0.13 [0.05]	6.56 [2.15]	0.43 [0.15]
hml	2.21 [0.69]	-3.19 [-0.92]	5.21 [1.61]	-0.29 [-0.09]	1.36 [0.41]	2.92 [0.84]	0.41 [0.12]
rmw	-8.79 [-2.32]	-3.06 [-0.83]	-3.97 [-1.12]	7.41 [2.26]	7.27 [2.15]	6.05 [1.46]	-0.29 [-0.06]
cma	9.37 [1.80]	13.15 [2.47]	12.80 [2.53]	0.89 [0.15]	14.35 [2.36]	12.74 [2.24]	-13.88 [-2.10]
umd	1.93 [1.16]	4.45 [2.63]	1.34 [0.80]	2.73 [1.63]	3.50 [2.04]	3.93 [2.18]	2.69 [1.55]
# months	728	728	728	732	732	630	626
$\bar{R}^2(\%)$	16	13	16	15	10	8	18

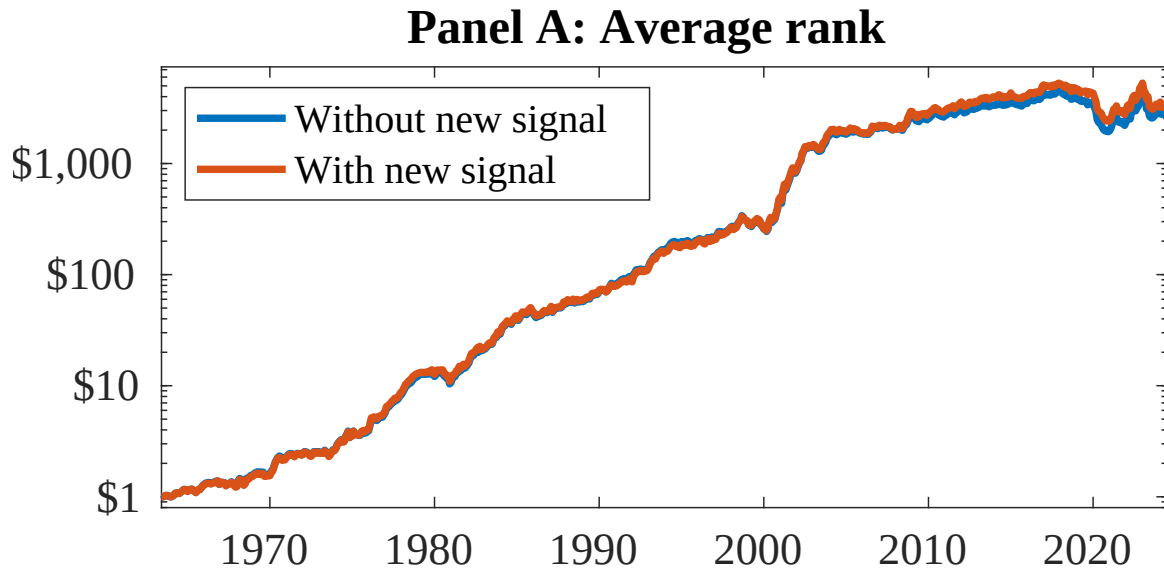


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EFI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

- Baker, M. and Wurgler, J. (2000). The equity share in new issues and aggregate stock returns. *Journal of Finance*, 55(5):2219–2257.
- Baker, M. and Wurgler, J. (2002). Market timing and capital structure. *Journal of Finance*, 57(1):1–32.
- Barber, B. M. and Odean, T. (2008). All that glitters: The effect of attention and news on the buying behavior of individual and institutional investors. *Review of Financial Studies*, 21(2):785–818.
- Boudoukh, J., Michaely, R., Richardson, M., and Roberts, M. R. (2007). On the importance of measuring payout yield: Implications for empirical asset pricing. *Journal of Finance*, 62(2):877–915.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Daniel, K. and Titman, S. (2006). Market reactions to tangible and intangible information. *Journal of Finance*, 61(4):1605–1643.
- DellaVigna, S. and Pollet, J. M. (2009). Investor inattention and friday earnings announcements. *Journal of Finance*, 64(2):709–749.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.

- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Hirshleifer, D. and Teoh, S. H. (2003). Limited attention, information disclosure, and financial reporting. *Journal of Accounting and Economics*, 36(1-3):337–386.
- Hong, H., Lim, T., and Stein, J. C. (2000). Bad news travels slowly: Size, analyst coverage, and the profitability of momentum strategies. *Journal of Finance*, 55(1):265–295.
- Hong, H. and Stein, J. C. (1999). A unified theory of underreaction, momentum trading, and overreaction in asset markets. *Journal of Finance*, 54(6):2143–2184.
- Hou, K., Xue, C., and Zhang, L. (2020). Replicating anomalies. *Review of Financial Studies*, 33(5):2019–2133.
- McLean, R. D. and Pontiff, J. (2016). Does academic research destroy stock return predictability? *Journal of Finance*, 71(1):5–32.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.
- Peng, L. and Xiong, W. (2006). Investor attention, overconfidence and category learning. *Journal of Financial Economics*, 80(3):563–602.